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Roll No

EE-5005(3)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Communication Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Give classification of signals. How can you differentiate deterministic and random signals? 7
b) State and prove the following properties of Fourier transform. 7
 - i) Time shifting
 - ii) Time convolution
2. a) Find the Fourier Transform of the following and draw the spectrum: 7
 - i) $e^{-bt} u(t)$
 - ii) $\cos \omega_0 t u(t)$b) What do you mean by modulation? Discuss the need for modulation. 7
3. a) Discuss the steps in PCM with the help of block diagram. What do you mean by companding? Explain. 7
b) Explain the synchronous detection method of DSB-SC signals. Explain the effect of phase and frequency errors in synchronous detection. 7

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4. a) Explain various methods of generation of FM. 7
b) Compare various AM techniques. 7
5. a) With the help of block diagram, explain the working of superheterodyne receiver. 7
b) What do you mean by AGC? Explain its principle. 7
6. a) Discuss sampling theorem. Explain the generation of sampled signal and how the original signal is recovered from a sampled signal. 7
b) Briefly explain: 7
 - i) QPSK
 - ii) QAM
7. a) What is transponder? Explain the various sections of a transponder channel. 7
b) Define EIRP and derive the formula for it. 7
8. Write short notes on (any two) 14
 - a) Gaussian and Rayleigh probability density function
 - b) IF amplifiers
 - c) TDMA vs FDMA
 - d) Link Budget Calculations
